

GARVIT GUPTA

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PROFILE SUMMARY

Software Engineer specializing in AI/ML with strong foundations in Data Structures, and backend systems. Experienced in building end-to-end ML pipelines and integrating intelligent models into scalable applications. Skilled in Python, Scikit-learn, and data-driven problem solving with a focus on performance and clean architecture

TECHNICAL SKILLS

Languages & Frameworks: Python, Java, HTML5, CSS, React.js, Node.js, Express.js, FastAPI

Machine Learning: Scikit-learn, Pandas, NumPy, Matplotlib, Feature Engineering, NLP, Computer Vision

Databases & Core CS: MongoDB, MySQL, Data Structures & Algorithms, OOP

Backend & Security: RESTful APIs, JWT Authentication, RBAC, MVC Architecture

Tools & Platforms: Git, GitHub, VS Code, Docker, Postman

PROJECTS

Walmart Sales Performance Analytics | *Python, Pandas, NumPy, Scikit-learn, Matplotlib* | [🔗 Link](#)

- Built an end-to-end **machine learning pipeline** to predict Walmart sales using **Linear Regression** and **Random Forest** on **421K+** retail records
- Performed **data preprocessing and feature engineering** on multi-source datasets (sales, stores, features) using **Pandas** and **NumPy**
- Achieved **97.07% prediction accuracy** with Random Forest Regressor through **hyperparameter tuning** and feature optimization
- Applied **backward feature elimination** to identify top 6 predictive features, reducing model complexity by **40%**

Smart Product Pricing Prediction System | *Scikit-learn, Pandas, NumPy, NLP* | [🔗 Link](#)

- Built end-to-end **ML pricing model** for **75K+ e-commerce products** using multimodal features from text descriptions and product images
- Implemented **NLP pipeline** with TF-IDF vectorization and text preprocessing to extract semantic features from product descriptions
- Optimized model performance achieving **SMAPE score of 48.4** through feature engineering and hyperparameter tuning

AI-Scalable URL Shortener | *Node.js, Express.js, React.js, MongoDB, Redis, Docker, FastAPI* | [🔗 Link](#)

- Architected a scalable full-stack URL shortener using **Node.js**, **Express.js**, and **React.js** with a 6-service microservices architecture containerized via **Docker Compose** for consistent cross-environment deployment
- Integrated a **FastAPI** microservice with **Scikit-learn** Random Forest classifier to detect and block malicious URLs in real time before shortening
- Secured REST APIs using **Helmet**, **express-rate-limiter**, **express-mongo-sanitize**, and **xss-clean** against DDoS, XSS, and NoSQLinjection attacks

EDUCATION

Vellore Institute of Technology

Bachelor's in CSE Specialization in Artificial Intelligence & Machine learning

Nov. 2022 – Nov. 2026

CGPA : 7.99

CERTIFICATIONS & ACHIEVEMENTS

- IBM GEN AI Using IBM Watsonx
- Applied Machine Learning in Python (Coursera)
- Solved 150+ DSA problems, strengthening algorithmic thinking and problem-solving skills.
- Represented college in 5+ inter-college cricket tournaments, developing teamwork and collaboration.